

AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer specialized in representation learning, Graph Neural Networks (GNN), and Transformer models. Passionate about Explainable AI (XAI) and modeling complex similarity relationships from multimodal data. PhD applicant for clinical reasoning support via the construction of patient graphs and evaluation of representativeness/atypicality.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, **Metric Learning**, **Autoencoders**

AI Techniques: GNN, Transformers, Latent Embeddings, **Explainable AI (XAI)**, Multi-modal Learning

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

Soft Skills: Technical leadership, Communication, Stakeholder management

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Digital Twin & Predictive Maintenance 2026

Sonatrach — Algeria

Developed a Digital Twin for gas turbines enabling real-time monitoring and predictive maintenance. Implemented advanced anomaly detection and RUL estimation using deep learning models.

Network Engineering Intern Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA) 2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Informatics

KEY ACHIEVEMENTS

- Developed custom Transformer models (BERT, GPT, RAG) for real-world multilingual applications.
- Built Explainable AI (XAI) platforms for HR tech and public institutions — prioritizing rea-

soning transparency.

- Learned graph representations over 60,000 nodes (+24% accuracy in similarity matching).
- Delivered complete ML pipelines: multimodal data ingestion, validation, training, and monitoring.
- Designed human-in-the-loop systems reducing bias and drift by 32%.

PROJECTS

Graph-Based Similarity Intelligence (Career Connect)

2024–2025

Modeled complex similarity relationships using **GNN**. Learned latent representations for robust semantic matching and identification of atypical profiles.

Tech: PyTorch Geometric, SBERT, FastAPI — Concepts: GNN, Relational Embeddings, Metric Learning

Recognizini — Latent Representation & Feedback

2025

Generated and updated latent embeddings with an active feedback loop to correct the model and identify outlier cases. A direct parallel to interactive clinical reasoning.

Tech: OpenCV, PyTorch, FastAPI — Concepts: Embeddings, Anomaly Detection, Label Propagation

Multimodal NLP Pipeline with Multi-Optimization

2025

Classification architecture optimized via PSO and Genetic Algorithm. Managed robustness and stability of classification results in complex texts.

Tech: DistilBERT, PyTorch, FastAPI — Concepts: Multimodal Models, Optimization, Robustness

Mini-GPT (Decoder-only)

2025

Experimental implementation of a compact GPT to understand sequential generation and LLM stability. Advanced hands-on experience with attention mechanisms.

Tech: PyTorch, CUDA, AdamW — Concepts: Transformers, Language Models, Representation Learning

North African Sentiment BERT

2025

LoRA adaptation of BERT for analyzing highly noisy and non-standardized textual data.

Tech: PyTorch, LoRA, MLflow — Concepts: Fine-tuning, LLM, XAI

Geo-RAG & Augmented Generation

2025

RAG system integrating structured (graphs, geo) and unstructured (text) data for explainable reasoning.

Tech: ChromaDB, LangChain — Concepts: RAG, Text Mining, Explainability

Elliptic++: Graph Anomaly Detection

2025

Identified fraudulent "cases" in massive transaction networks (200k+ nodes) using graph autoencoders.

Tech: PyTorch Geometric, NetworkX — Concepts: GNN, Atypicality Detection

LANGUAGES

Arabic: Mother Tongue | English: Advanced | French: Conversational